

Introduction

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Applied Quantitative Methods II
MA in Social Sciences, Spring 2026

Course overview

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- Less theory, more hands-on work with data
- Goal: go from research question to answer

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- How to interpret and visualize model results
- How to evaluate whether a model is appropriate
- How to work with different types of data (panel, spatial, etc.)
- Best practices in computing and reproducibility

Course structure

Feb 5	Introduction
Feb 12	Applied regression
Feb 19	Applied regression II (binary)
Feb 26	Interpretation and diagnostics
Mar 5	Best practices in computing
Mar 12	Panel data I
Mar 19	Panel data II
Mar 26	Spatial data
<i>Easter break</i>	
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Apr 16	Other outcomes
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Apr 30	Exam + Review

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Evaluation

- Problem sets (20%)
 - Started in class, finished at home
 - Short deadlines
- Proposal presentation and peer review (10% + 10%)
- Final essay (30%)
 - Small research note (max 3,000 words)
 - Original data analysis using R
- Exam (30%)

Roadmap

The Big Picture

Workflow Basics

The research process

Theory $\longleftrightarrow$ Data Generating Process $\longleftrightarrow$ Data

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- Our research strategy connects theory to data

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- Common mistake: picking a method and then looking for a question
- In this course: we learn tools, but always ask *why this tool for this question?*

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- We use statistical models to make inferences about it

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- Statistical models help us deal with this noise

The logic of statistical inference

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- We're doing the reverse: from data back to process

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Learning to use computers as tools

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- We'll have a session on computing, project management, etc – but today, some notes on version control

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- How do you collaborate without overwriting each other's work?

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- Multiple people can work simultaneously

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- Many journals now require or encourage sharing code via GitHub

Git and GitHub

Git

- A version control system
- Runs locally on your computer
- Tracks changes to files

GitHub

- A web platform that hosts Git repositories
- Stores your code online
- Enables sharing and collaboration

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4. **Push** your commits to GitHub
 - Upload your local changes to the cloud

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- All do the same thing—choose what works for you

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 - Detailed instructions in the assignment document

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- Appropriate data for the question
- Right statistical model for the data
- Correct interpretation of results
- Honest about limitations and uncertainty

Looking ahead

- Next session: Applied regression

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- Regression as conditional expectations

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- Multiple regression and control variables
- Interaction effects and presenting results

For next week

- Check readings if needed
- Review your notes on OLS from AQMSS-I
- **Finish Assignment 1**

Questions?